

Sadir Ahmed Zidan

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Research Interests

- Large Language Models: fine-tuning, agentic systems, retrieval-augmented generation
- Computer Vision: monocular depth estimation, 3D reconstruction, document/scene understanding
- Efficient/Edge Deep Learning: model compression, quantization, low-resource deployment
- Applied Explainable AI (XAI) for healthcare and safety-critical prediction tasks

Education

BSc in Computer Science & Engineering

American International University Bangladesh (AIUB) | January 2020 - August 2024

- CGPA: 3.77/4.00
- Thesis: "A Study on the Use of Machine Learning Algorithms for Predicting Mental Health Conditions"

Research Experience

Five Years of VAERS: Explainable Machine Learning for COVID-19 Vaccine Adverse Event Severity and Symptom Co-occurrence

Co-authored | 2026 – Present

- Built a dual-model explainable ML system on 1,052,335 VAERS COVID-19 adverse event records (2019–2026): an XGBoost severity classifier (five-level scale) achieving 0.8498 AUC-ROC and 0.81 Fatal-class recall, and a neural network symptom co-occurrence model reaching 93.79% Top-3 accuracy.
- Applied SHAP and LIME to produce patient-level explanations, forming a transparent five-step clinical narrative for post-vaccination safety surveillance.
- Manuscript under review at AIUB Journal of Science and Engineering (AJSE), following a full peer-review and resubmission cycle.

NSSLDepth : Lightweight Self-Supervised Monocular Depth Estimation for Edge Devices (working title)

Nasir Syntax Solution Limited (concern of Nasir Group) | 2025 – Present

- Designing a self-supervised monocular depth estimation network replacing the standard ResNet-18 encoder with a MobileNetV2 backbone, targeting real-time inference on mobile and edge hardware.
- Developing a custom "Mobile-Aware Depth Decoder" using cross-gated feature fusion and learned scale calibration to balance accuracy and latency.
- Built evaluation pipelines on the KITTI benchmark and deployment tests across Android and iOS devices via TensorRT/CoreML conversion; manuscript in preparation.

A Study on the Use of Machine Learning Algorithms for Predicting Mental Health Conditions (undergraduate thesis)

American International University Bangladesh | 2023

- Evaluated classical ML algorithms (Logistic Regression, kNN, Random Forest, SVM, Stacking) for classifying mental health conditions from structured survey data.
- Found Stacking and Random Forest outperformed other models across accuracy, recall, precision, and F1.
- Completed, unpublished.

Publications

1. K. I. Rushee, M. A. Nadif, and S. A. Zidan. "Five Years of VAERS: Explainable Machine Learning for COVID-19 Vaccine Adverse Event Severity and Symptom Co-occurrence." AIUB Journal of Science and Engineering (AJSE).
 - Under review, 2026 (resubmitted following peer review).
2. T. Kamran, M. A. Nadif, S. A. Zidan, and M. Sarwar. "NSSLDepth: A Lightweight Self-Supervised Monocular Depth Estimation Network for Edge and Mobile Deployment" (working title).
 - Manuscript in preparation, 2025-present. Nasir Syntax Solution Limited.

Work Experience

Machine Learning Engineer - Fulltime, Onsite

Nasir Syntax Solution Limited (concern of Nasir Group), January 2024 - Present

- Designed and shipped production RAG and agentic LLM systems using LangChain, FAISS, Milvus, and custom retrieval pipelines for vector search, personalized search and custom chatbots. reduced retrieval latency by 40% through pipeline optimization and index tuning.
- Fine-tuned and quantized LLMs with LoRA/QLoRA, built dynamic model loading and inference optimization workflows, achieved 60% reduction in model memory footprint enabling deployment on resource-constrained and edge-compatible environments.
- Developed computer vision pipelines including object detection, segmentation, ocr for document data extraction and 3D reconstruction projects.
- Trained custom deep learning models from scratch and integrated pretrained models.
- Built and deployed scalable FastAPI inference services with Docker, MLflow tracking, request validation, rate-limiting, and monitoring.

AI Developer - Part time, Remote

GradMate.ai , Apr 2025 - Nov 2025

- Built high-performance agentic RAG systems and multi-step AI workflows using LangChain and Groq's ultra-fast LLM inference.
- Implemented real-time speech-to-text (STT) and text-to-speech (TTS) integrations into educational AI agents.
- Optimized latency and cost for Groq/LangChain pipelines serving live user-facing features, cut average inference response time by 35% and reduced per-query API cost through prompt compression and model routing strategies.
- Collaborated on modular AI architecture, data contracts, and retrieval quality standards with frontend and platform teams.

AI Engineer - Contractual, Remote

Ambitionix, May 2025 - Feb 2026

- Fine-tuned LLMs for specific tasks.
- Created agents and APIs for deployment.
- Collaborated with cross-functional teams to achieve project goals.

Machine Learning Intern

Nasir Syntax Solution Limited (concern of Nasir Group), October 2023 - December 2023

- Conducted research on personalized recommendation systems, chatbots, and text generation using LLMs and computer vision.
- Scraped and preprocessed data, including tokenization and vectorization, and fine-tuned LLMs for specific applications.

Skills

- **Programming Languages:** Python, R
- **Frameworks & Libraries:** TensorFlow, Keras, LangChain, Groq, PyTorch, Fast API, Pandas, Faiss, Llama cpp, ONNX, Ollama, Hugging Face, Streamlit
- **Tools & Platforms:** Docker, MLFlow, Kubernetes, GitHub, VS Code, R Studio, Colab, LoRA
- **Databases:** SQL, MongoDB
- **Operating Systems:** Windows, Linux

Personal Projects

- **Multilingual RAG System for Bengali and English Queries**
 - A system to process queries in English and Bengali, retrieve relevant information from a PDF document (specifically the HSC26 Bangla 1st Paper book), and generate meaningful answers.
- **Face Mask Detection**
 - A system to detect facial masks using a cascade classifier. Used a large dataset that varies with different types of poses with masks worn by humans. The model got good accuracy after implementation.
- **Drowsiness Detection**
 - A real-time system using machine learning algorithms and image processing to detect driver drowsiness. It can continuously monitor the driver while they are driving and provide immediate feedback or alerts.
- **Sentiment Analysis Chatbot**
 - This project demonstrates a simple sentiment analysis chatbot that analyzes the user's mood based on their input and provides contextual responses.
- **Life Insurance Support Assistant**
 - Agentic life-insurance assistant built using LangChain + Groq, designed to answer domain-specific questions about policy types, benefits, eligibility, and claims, using structured category-based reasoning.
- **Voice-Enabled Agentic RAG**
 - A modular, local Retrieval-Augmented Generation (RAG) pipeline for multi-domain document Q&A, featuring PDF cleaning, adaptive chunking, FAISS vector indexing, multi-agent retrieval, and both text and voice CLI interfaces.

References

Mohammad Abu Nadif
Machine Learning Engineer & Researcher
Nasir Syntax Solution Limited
Email: nadif@syntaxbd.com
Relationship: Team Lead and Research Supervisor